SUMMER RESEARCH INTERNSHIP REPORT

Finite Element Analysis of Reinforced Concrete Beam Under Four-point Bending

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Four-point bending simulation:

1.1. Part and assembly:

A three-dimensional finite element model of a reinforced concrete (RC) beam was developed in ABAQUS to simulate the four-point bending test. The model consisted of a concrete beam, longitudinal reinforcement bars(rebars), transverse reinforcement (stirrups), and supports.

The concrete beam was modelled as a solid component, while longitudinal reinforcement and stirrups were modelled separately and embedded within the concrete beam using the embedded region constraint. The longitudinal reinforcement was provided along the length of the beam to resist flexural stresses, whereas stirrups were placed at specified spacing to enhance shear resistance and confinement of the concrete.

The loading system consisted of two loading plates positioned symmetrically about the beam centreline and two support plates near the beam ends. The assembly was configured to represent the actual four-point bending test setup.

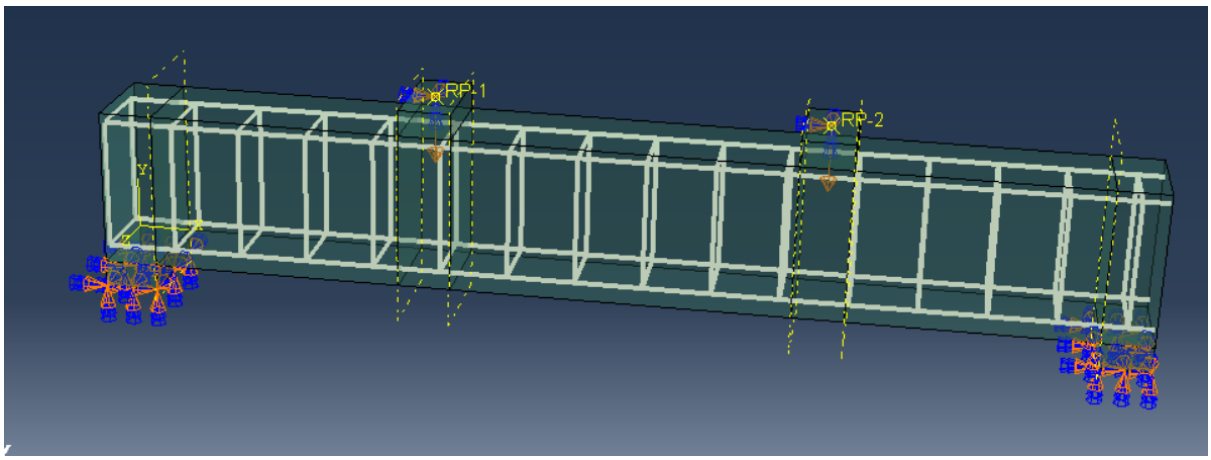


Fig1.2.1 Finite element assembly of the reinforced concrete beam showing concrete beam longitudinal reinforcement, stirrups, loading plates and support plates.

1.2. Material properties:

Concrete beam is assigned as cement-concrete material which has density 2.4E-09 ton/mm^3 and for linear elastic material – Young's modulus and Poisson's ration are taken as 30000Mpa and 0.2 respectively. The longitudinal rebars and stirrups are assigned as steel which has density 7.85E-09 ton/mm^3 and for linear elastic properties - Young's modulus and Poisson's ration are taken as 200000Mpa and 0.3.

1.3. Element Type and Meshing:

The beam and support components were discretised using eight-node linear brick elements with reduced integration (C3D8R). A finer mesh was adopted in the vicinity of the loading

and support regions to accurately capture stress concentrations and crack propagation, while a relatively coarser mesh was used elsewhere to reduce computational cost.

1.4. Interactions and Constraints:

Interactions between the beam and support were defined to ensure proper transfer of load and boundary conditions. Tie constraints were assigned between the supports and the beam to establish a bonded connection. An embedded region constraint was used to model the reinforcement embedded within the concrete beam. Coupling constraints were applied at the loading locations to distribute the applied load uniformly across the loading plates.

1.5. Boundary conditions and loading:

The beam was simply supported at both ends through support plates. The movement of supports were restricted in all three directions to prevent rigid body motion while applying the vertical load. Vertical loads were applied downward on the two top loading plates to stimulate the four-point bending test configuration. The loads were positioned symmetrically about the beam centreline, creating a constant bending moment region between the two loading points. The analysis was performed under static loading conditions using ABAQUS. The applied loading arrangement enabled the evaluation of beam flexural behaviour, displacement response, and stress distribution under bending.

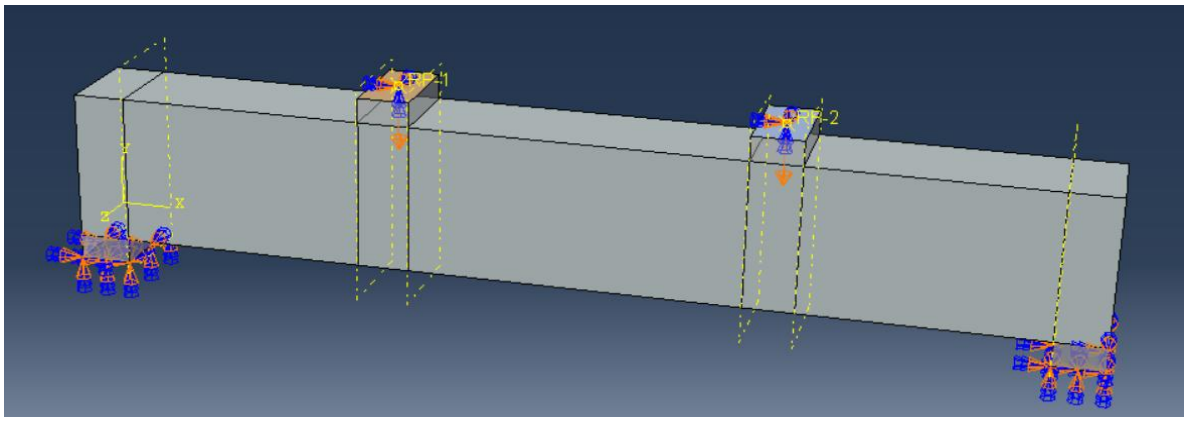


Fig 1.2.2. Boundary conditions and loading arrangement for the four-point bending test

1.6. Results and Discussion:

The displacement contour obtained from the analysis is shown in figure 6. Maximum deflection occurred at the mid-span region of the beam, where bending effects were most significant. The deformation pattern confirms the expected flexural behaviour under four-point loading.

The stress counter (S22) obtained from ABAQUS is shown in figure 7. Higher tensile stressed developed in the bottom portion of the beam near the constant moment region, while compressive stresses were observed near the top surface. The stress distribution is consistent with the theoretical behaviour of beams subjected to flexural loading.

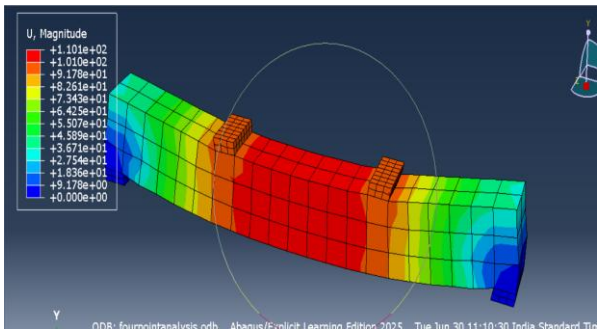


Fig 1.2.3. Displacement contour of the beam under four-point bending.

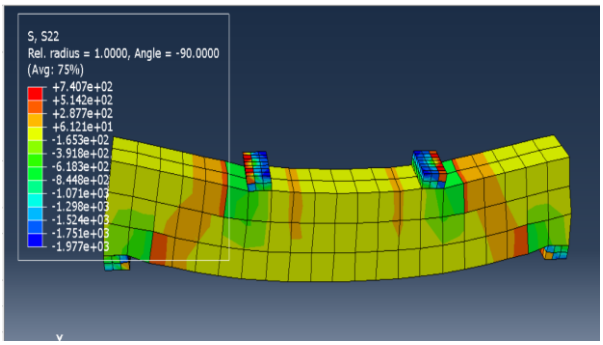


Fig 1.2.4. Stress distribution in the concrete beam under four-point bending.

1.7. Conclusion:

A three-dimensional finite element model of a reinforced concrete beam was developed in ABAQUS and analysed under four-point bending loading. The results showed maximum deflection at the beam mid-span and significant stress concentrations in constant moment region. The study demonstrated the capability of ABAQUS to effectively predict the flexural behaviour and structural response of reinforced concrete beams.